

AZEEM EHTISHAM

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Software and systems engineer building real-time telemetry, edge, and interactive 3D applications. Designs reliable software through deterministic fallbacks, hands-on validation, and clear user outcomes.

EDUCATION

Purdue University

Bachelor of Science, Computer Information Technology | GPA 3.60

West Lafayette, IN

May 2024 – May 2028

SOFTWARE AND SYSTEMS EXPERIENCE

Undergraduate Research Fellow / OUR Scholar

May 2026 – Present

Purdue SURF / OUR Scholars Program

West Lafayette, IN

- Engineered a **Meta Quest 3** mixed-reality training system in **Unity 6.2** and **C#**, placing a full-scale industrial bioreactor into real space through low-latency passthrough rendering
- Designed real-time state synchronization for **up to 5 headsets** with **Unity Relay** and replicated **NetworkVariables**, ensuring every user receives the same selected-component and session state
- Built **Unity Editor** tools that convert raw FBX geometry into wired hotspots and colliders, reducing **38 assets** from multi-step setup to one-click generation and validating each on-device

NASA SUITS Team Lead

Sep 2025 – Present

SEARCH

West Lafayette, IN

- Directed live system testing for **NASA design evaluators**, coordinating real-time demonstrations across subteams to expose integration issues and validate AR features under field conditions
- Deployed a **Raspberry Pi 5** edge node with a local Wi-Fi access point, SSH, and VNC, then resolved DNS and IPv6 failures to restore reliable remote access for the development team

Team Lead / Scrum Master

Jan 2026 – Jul 2026

The Data Mine - U.S. Space Force

West Lafayette, IN

- Promoted to lead **12 students** after a high-performance evaluation, translating mission objectives into technical milestones that maintained **100% completion** across **3 cyber-defense workstreams**

SELECTED SOFTWARE PROJECTS

Cabin Telemetry Dashboard | PHP, MySQL, Chart.js, HTML5

- Built a **telemetry dashboard** for a remote family cabin, organizing **6 solar, battery, environmental, load, heater, and camera feeds** into an at-a-glance interface for a nontechnical owner
- Replaced **8 separate full-history requests every 5 seconds** with one consolidated current-state endpoint and bounded history queries, eliminating repeated full-table transfers as sensor data grows
- Modernized **28 PHP endpoints** with parameterized **MySQL** queries, eliminating deprecated database calls and closing **6 identified injection paths** while preserving the existing microcontroller data contract

Bar Babes | React Native, FastAPI, MongoDB, NFC, Sensor Telemetry

- Built a real-time sensor-to-decision pipeline and **FastAPI/MongoDB** backend that converts accelerometer and reaction data into impairment scores while synchronizing geolocation and **NFC** events
- Preserved safety-critical output with a deterministic fallback when the AI service is unavailable; shipped the hardware-connected system with a team of **4 in under 36 hours**

VibeCheck | FastAPI, Python, WebSockets, SQLAlchemy, Docker

- Engineered a production security-scanning platform with an asynchronous **FastAPI, SQLAlchemy, and Pydantic v2** backend, deployed at **aedify.ai**
- Validated health, assessment, pagination, and search behavior with a **21-test pytest suite**, achieving a **100% pass rate** across all tested endpoints

SKILLS

Languages & Tools: C#, C++, Python, JavaScript/TypeScript, PHP, MySQL, Chart.js, Unity 6.2, FastAPI, Docker, pytest, Git/GitHub, Linux

Software Systems: Real-time client/server synchronization, REST APIs, WebSockets, telemetry dashboards, embedded and edge systems, networking, software testing, Raspberry Pi 5, Meta Quest 3